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61.

Solution :

Given

$T = 400K ; \Delta H^0 = 77.2\ kJ\ mol^{-1} = 77200$

$J\ mol^{-1} ; \Delta S^0 = 122\ JK^{-1}\ mol^{-1}$

$\Delta G^0 = -2.303\ RT \log K_{eq}$

$$\log K_{eq} = \frac{\Delta G^0}{2.303\ RT}$$

$$\log K_{eq} = -\frac{\left(\Delta H^0 - T\Delta S^0\right)}{2.303\ RT}$$

$$\log K_{eq} = -\left(\frac{77200 - 400 \times 122}{2.303 \times 8.314 \times 400}\right)$$

$$\log K_{eq} = -\left(\frac{28400}{7659}\right)$$

$\log K_{eq} = -3.7080$

$K_{eq} = \text{antilog}\ (-3.7080)$

$K_{eq} = 1.95 \times 10^{-4}$

62.

Solution :

Given

$T = 298K ; \Delta U = -742.4\ kJ\ mol^{-1}$

$\Delta H = ?$

$\Delta H = \Delta U + \Delta n_g\ RT$

$\Delta H = \Delta U + (n_p - n_r)\ RT$

$$\Delta H = -742.4 + \left(2 - \frac{3}{2}\right) \times 8.314 \times 10^{-3} \times 298$$

$= -742.4 + (0.5 \times 8.314 \times 10^{-3} \times 298)$

$= -742.4 + 1.24$

$= -741.16\ kJ\ mol^{-1}$

63.

Solution :

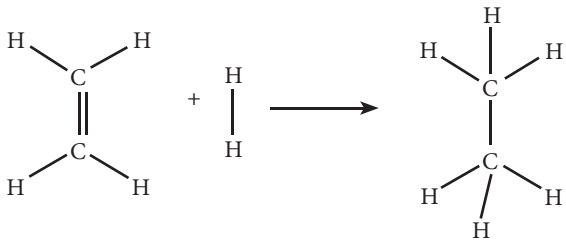
Given :

$E_{C-H} = 414\ kJ\ mol^{-1}$

$E_{C-C} = 347\ kJ\ mol^{-1}$

$E_{C=C} = 618\ kJ\ mol^{-1}$

$E_{H-H} = 435\ kJ\ mol^{-1}$



Ethylene Ethylene

$\Delta H_r = \Sigma\ (\text{Bond energy})_r - \Sigma\ (\text{Bond energy})_p$

$\Delta H_r = (E_{C=C} + 4E_{C-H} + E_{H-H}) - (E_{C-C} + 6E_{C-H})$

$\Delta H_r = (618 + (4 \times 414) + 435) - (347 + (6 \times 414))$

$\Delta H_r = 2709 - 2831$

$\Delta H_r = -122\ kJ\ mol^{-1}$